SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Deemed to be University) • Chennai

ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

LAB MANUAL

EXPERIMENT NO 4:

Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

12.5 ps Gaussian pulse · 10 km SMF · −160 ps/nm ideal dispersion compensation

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Submission Date	10 October 2026

EXPERIMENT NO 4:

Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

AIM / OBJECTIVES

1. To demonstrate dispersion compensation in OptiSystem with the Ideal Dispersion Compensation (FBG) component.

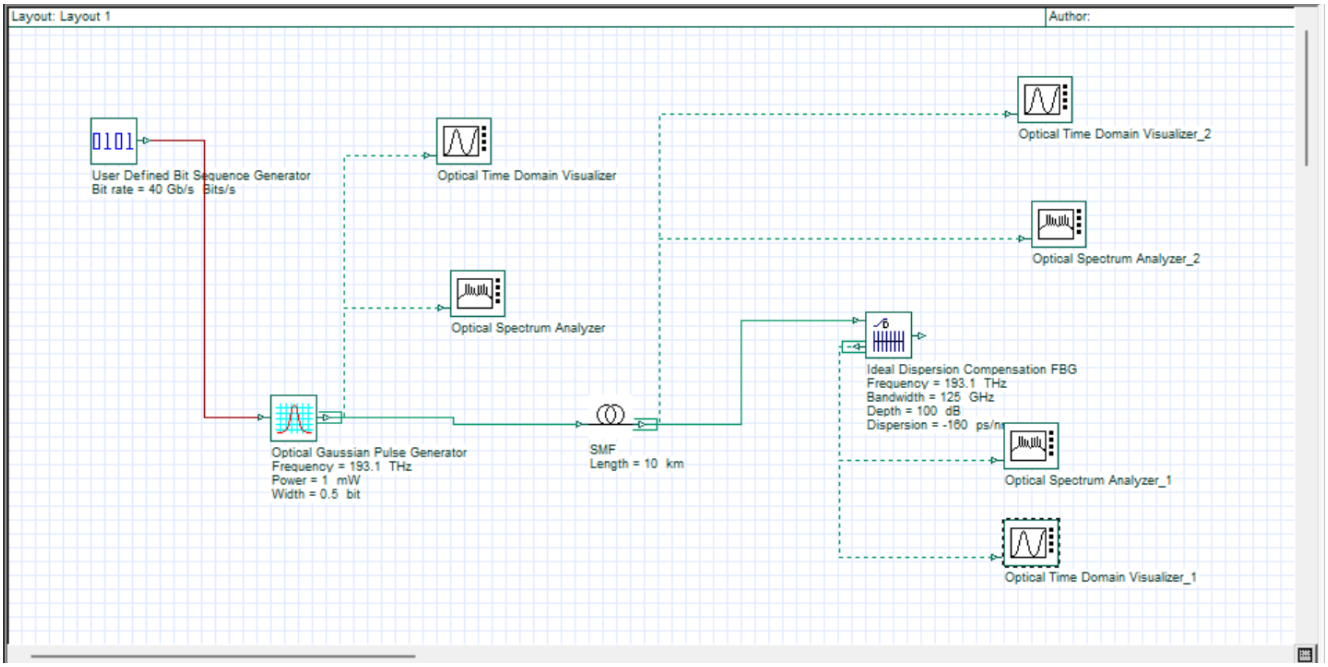
2. To observe how a 12.5 ps optical Gaussian pulse broadens after travelling through 10 km of single-mode fiber (SMF).

3. To cancel the accumulated dispersion of 160 ps/nm and confirm that the original pulse shape is recovered.

SOFTWARE / HARDWARE REQUIRED

• OptiSystem software

BLOCK DIAGRAM



OUTPUTS

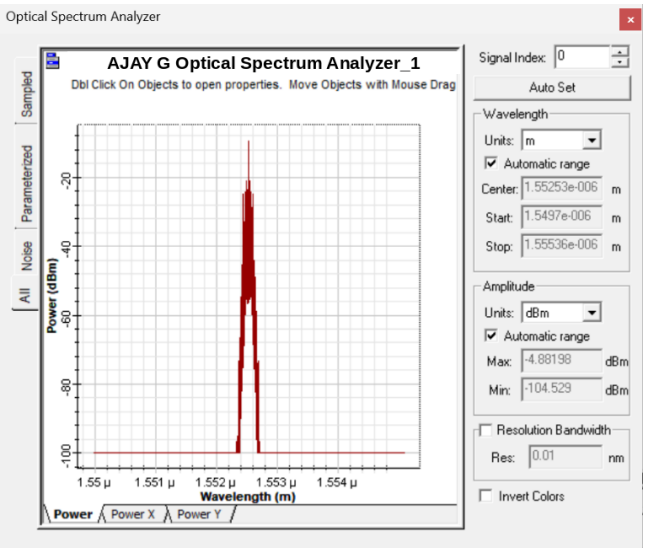


Fig. 1: AJAY G Optical Spectrum Analyzer_1

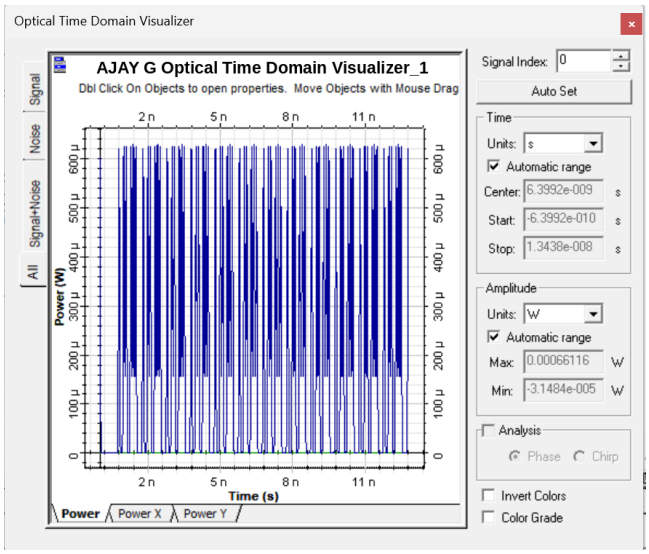


Fig. 2: AJAY G Optical Time Domain Visualizer_1

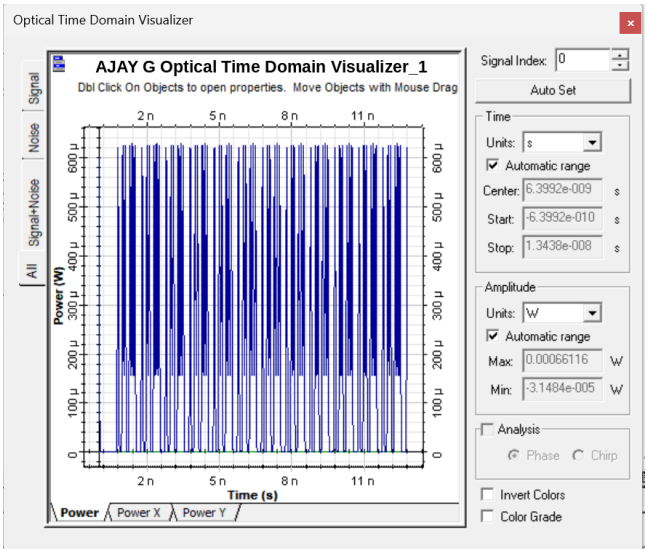


Fig. 3: AJAY G Optical Time Domain Visualizer_1 (compensated output)

REFLECTION

Running this simulation in OptiSystem showed me clearly how dispersion compensation reduces pulse spreading and restores signal quality in an optical fiber link.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, –160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of settings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

CONCLUSION

The Ideal Dispersion Compensation component in OptiSystem was used to demonstrate dispersion compensation successfully. After 10 km of SMF (accumulated dispersion of 160 ps/nm), the 12.5 ps Gaussian pulse spread to roughly four times its original width. Setting the compensator dispersion to –160 ps/nm brought the pulse back to its original shape, giving exact cancellation of the accumulated dispersion.